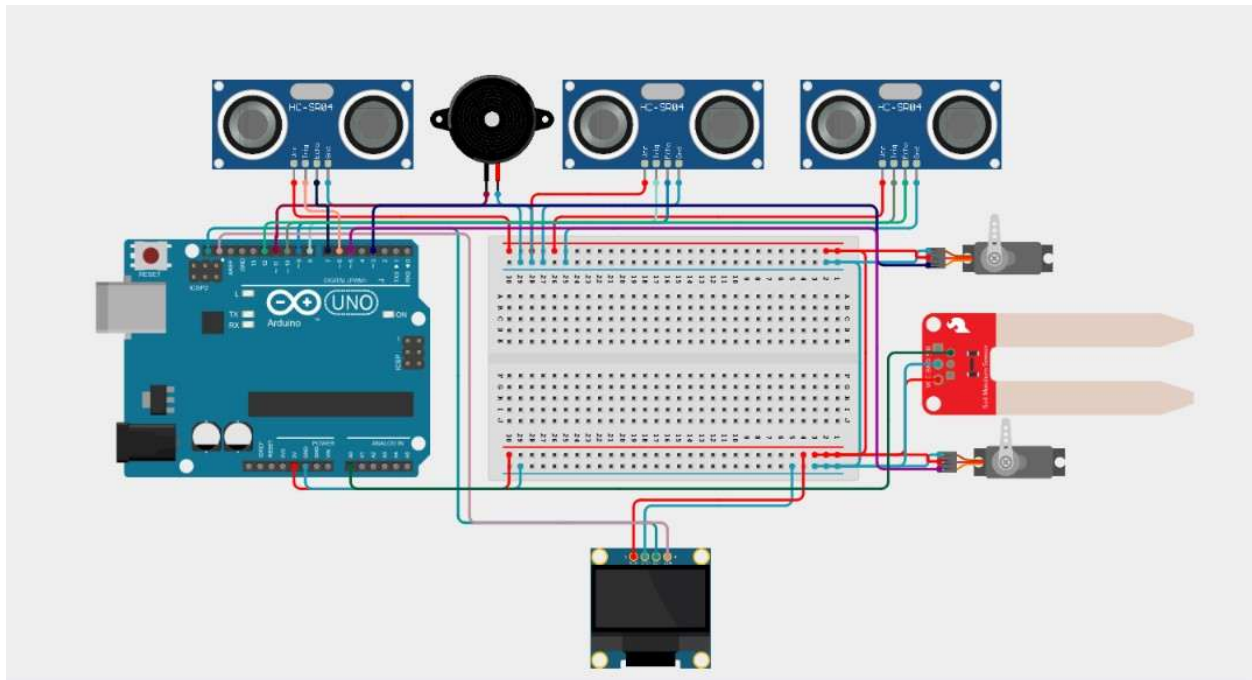


SMART DUSTBIN

Automatic Wet/Dry Waste Segregation with OLED-Based Real-Time Monitoring



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1. Introduction

Municipal and household waste management is one of the most persistent challenges faced by growing urban populations. A large fraction of the inefficiency in existing waste collection systems does not come from a shortage of infrastructure, but from the very first step in the chain: the point at which waste is generated and disposed of. Conventional dustbins require physical contact to open, provide no separation between biodegradable (wet) and non-biodegradable (dry) waste, and give no indication of how full they are until they visibly overflow. Each of these shortcomings compounds the workload on downstream sorting and collection systems.

The Smart Dustbin project addresses these three problems directly using a low-cost, microcontroller-based embedded system built around the Arduino Uno. The system automatically opens its lid when a hand or object is detected nearby, removing the need for physical contact and improving hygiene. It then uses a soil moisture sensor to determine whether the deposited waste is wet or dry, and drives a servo-controlled flap to route the waste into the correct compartment without any manual sorting. A third ultrasonic sensor continuously monitors the fill level of the bin and raises an audible alert once a configurable threshold is crossed, so that the bin is emptied before it overflows. All of this status information is displayed in real time on a compact OLED screen, giving an at-a-glance summary of the bin's operating state.

This project was undertaken as part of an embedded systems and IoT engineering course, with the goal of building practical, portfolio-grade experience in sensor interfacing, actuator control, real-time decision logic, and display programming on the Arduino platform. The design was fully developed and validated through circuit simulation in Circuit design prior to any physical assembly, allowing the sensor thresholds, servo angles, and control logic to be tuned safely before hardware deployment.

1.1 Motivation

Improper waste segregation at the source is one of the biggest bottlenecks in effective recycling and composting programs. When wet and dry waste are mixed at the point of disposal, recyclable material is frequently contaminated and rendered unusable, and biodegradable waste that could be composted instead ends up in landfills. A device that performs this segregation automatically, at the moment of disposal, removes the dependency on user discipline and awareness, and provides a scalable building block for smart-city waste management infrastructure.

1.2 Objectives

- Design a touchless, ultrasonic-sensor-based lid mechanism to improve hygiene and convenience.
- Implement automatic wet/dry waste classification using a soil moisture sensor and route waste accordingly using a servo-actuated flap.
- Continuously monitor bin fill level using ultrasonic ranging and generate an audible alert when the bin approaches capacity.
- Provide real-time visual feedback of system status (lid state, waste type, fill percentage, and alerts) on an OLED display.
- Validate the complete design through circuit simulation before physical implementation, and document the system for portfolio and future development purposes.

1.3 Scope of the Project

The current implementation is built and validated entirely in a Circuit design circuit simulation using an Arduino Uno as the controller. The scope covers sensor interfacing (three HC-SR04 ultrasonic sensors and one capacitive/resistive

soil moisture sensor), dual-servo actuation for the lid and the sorting flap, buzzer-based alerting, and OLED-based status reporting. The project does not, in its current form, include wireless connectivity, cloud logging, or a physical enclosure -these are identified as future scope in Section 12.

2. Schematic Diagram

The schematic below shows the complete electrical connectivity of the Smart Dustbin system: three HC-SR04 ultrasonic sensors, two servo motors, a soil moisture sensor, a piezo buzzer, an Arduino Uno, and an I2C OLED display, along with the shared 5V and ground rails.

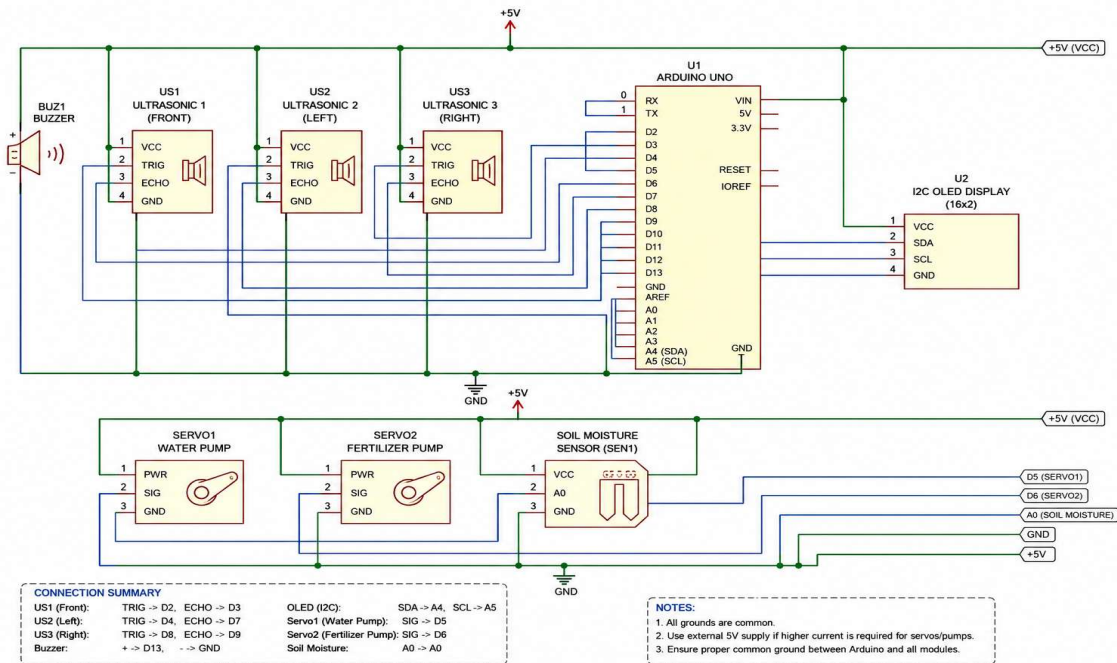


Fig. 2.1 -Complete system schematic showing sensor, actuator, and controller wiring

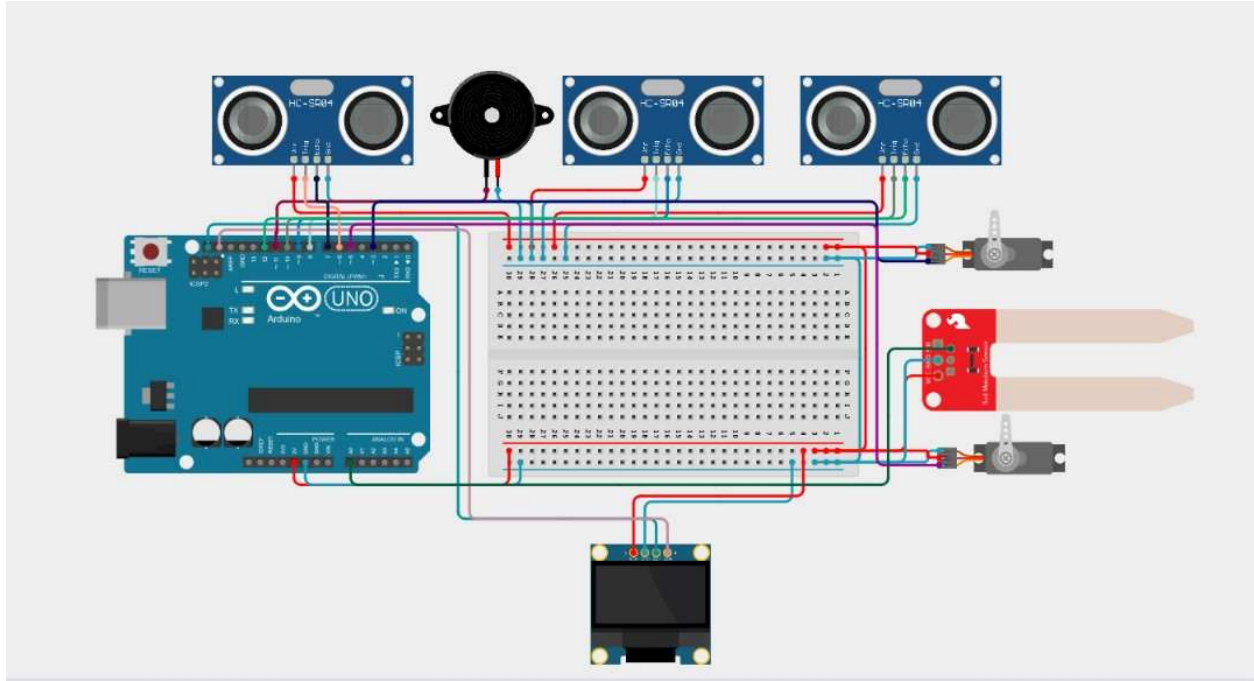
2.1 Pin Mapping Summary

Module	Function	Arduino Pin
US1 (Ultrasonic -Lid)	TRIG / ECHO	D6 / D7
US2 (Ultrasonic -Waste Detect)	TRIG / ECHO	D8 / D9
US3 (Ultrasonic -Fill Level)	TRIG / ECHO	D10 / D12
Servo 1 (Lid)	Signal	D3
Servo 2 (Sorting Flap)	Signal	D5
Soil Moisture Sensor	Analog Output	A0
Piezo Buzzer	Signal	D11
OLED Display (SSD1306, I2C)	SDA / SCL	A4 / A5

Note: an earlier revision of the schematic (used for initial planning) assigned different pin numbers to the ultrasonic sensors and used D5/D6 for the two servos. The pin mapping above reflects the final Arduino sketch and is the authoritative reference for hardware wiring.

3. Circuit Description

The circuit is built entirely around the Arduino Uno, which acts as the central controller reading all sensor inputs and driving all outputs. All modules share a common 5V supply rail and a common ground, which is essential for the ultrasonic sensors and the OLED display to operate reliably.



3.1 Ultrasonic Sensors (HC-SR04) -×3

Three identical HC-SR04 modules are used, each performing a distinct role. The first, mounted at the lid, detects the approach of a hand and triggers the lid-opening sequence. The second is positioned to detect when waste has actually been dropped into the collection chute, which gates the wet/dry classification logic so that it only runs when waste is present. The third is aimed down into the bin cavity to measure the distance to the surface of the accumulated waste, which is converted into a fill percentage. Each sensor's TRIG pin is driven by a digital output pin and its ECHO pin is read as a digital input, with distance computed from the round-trip time of the ultrasonic pulse.

3.2 Servo Motors -×2

Two hobby servo motors provide the mechanical actuation for the system. The first is mechanically coupled to the lid and swings between a closed position and a fully open position when triggered by the lid ultrasonic sensor. The second is coupled to a sorting flap positioned beneath the waste-detection sensor; it swings to one extreme to direct waste into the dry compartment and to the opposite extreme to direct waste into the wet compartment, based on the soil moisture reading taken at the moment waste is detected.

3.3 Soil Moisture Sensor

The soil moisture sensor is used here as a proxy for a wet/dry waste classifier -a low-cost and effective way of estimating moisture content in typical household waste. It outputs a variable analog voltage depending on the conductivity between its two probes, which is read on Arduino analog pin A0. A lower analog reading corresponds to higher moisture content; this reading is compared against a calibrated threshold to classify the deposited waste as wet or dry.

3.4 Piezo Buzzer

The buzzer is a simple two-terminal piezo element driven directly from a digital output pin. It is switched on continuously whenever the computed fill percentage of the bin equals or exceeds the configured full-bin threshold, and switched off otherwise, giving an audible warning that the bin needs to be emptied.

3.5 OLED Display Module

A 128×64-pixel SSD1306 OLED display communicates with the Arduino over the I2C bus (SDA/SCL), which keeps the wiring minimal -only four connections (VCC, GND, SDA, SCL) are required regardless of how much information is shown. The display is refreshed continuously with the current lid state, the most recently detected waste type, the fill percentage, and a full-bin warning when applicable.

3.6 Power Distribution

All modules are powered from the Arduino's regulated 5V pin during simulation and low-current bench testing. The schematic includes a note recommending an external 5V supply for the servo motors in a physical build, since servos can draw current spikes under load that may destabilize the Arduino's onboard regulator if too many peripherals are powered from it simultaneously. A common ground between the Arduino and every module is essential and is called out explicitly in the schematic notes.

4. Working Principle

The Smart Dustbin operates as a continuously looping real-time control system, re-evaluating all three sensing tasks -lid control, waste sorting, and fill monitoring -on every pass of the main loop, and updating the OLED display with the latest combined status after each pass.

4.1 Touchless Lid Operation

The lid ultrasonic sensor continuously measures the distance to any object in front of it. When this distance falls below a configured threshold (indicating a hand or object has approached the bin), the lid servo is commanded to its open angle and the system records the lid as open. Once the object moves away and the sensor no longer detects it within range, the system waits briefly to avoid premature closing before returning the lid servo to its closed position.

4.2 Wet/Dry Waste Detection and Sorting

The second ultrasonic sensor watches for waste actually falling past a fixed point inside the chute. When an object is detected within the drop-detection threshold, the system immediately takes a reading from the soil moisture sensor. If the analog value is below the configured wet threshold, the flap servo swings to the wet-waste angle, directing the item into the wet compartment; otherwise, it swings to the dry-waste angle. After a short delay to allow the waste to pass through, the flap automatically resets to its default resting position, ready for the next item.

4.3 Fill-Level Monitoring and Alerting

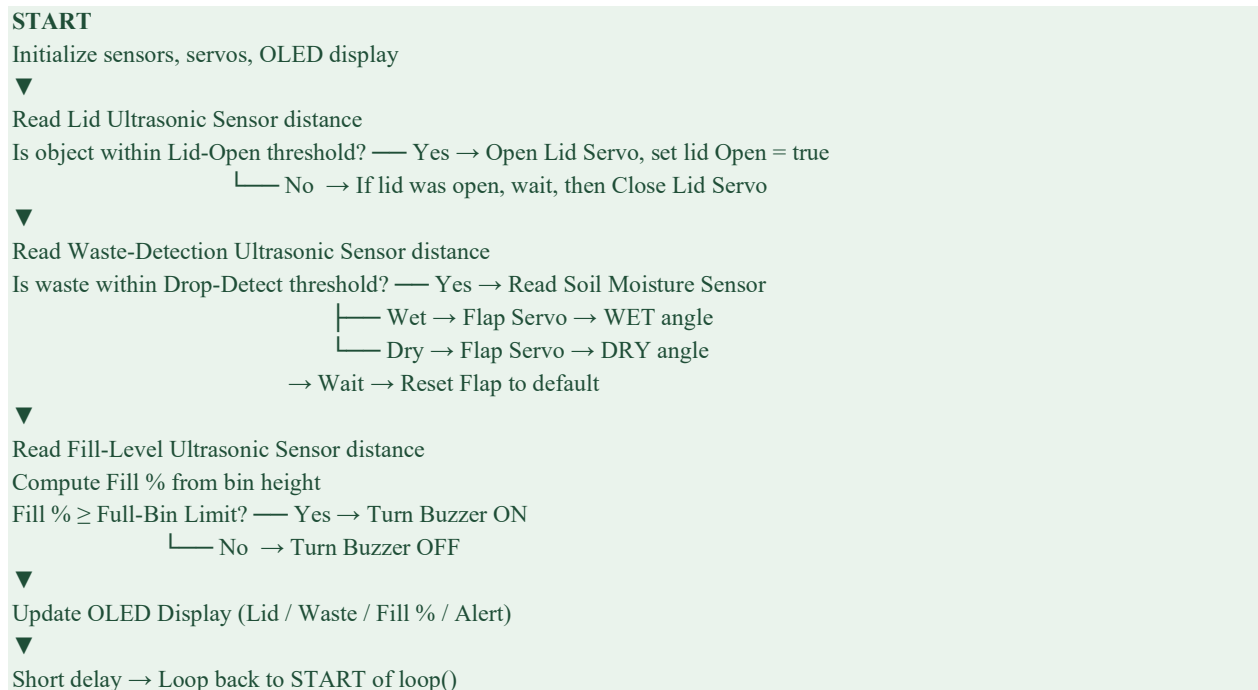
The third ultrasonic sensor measures the distance from the top of the bin down to the surface of the accumulated waste. This raw distance is mapped against the known internal height of the bin to compute a fill percentage between 0% and 100%. Whenever this percentage reaches or exceeds the configured full-bin limit, the buzzer is activated continuously and the OLED display shows a bin-full warning, prompting the bin to be emptied.

4.4 Status Reporting

On every iteration of the control loop, the current lid state, most recent waste classification, fill percentage, and full-bin status are written to the OLED display, giving continuous, real-time visibility into the condition of the bin without needing to open it or inspect it manually.

5. Flowchart

The flowchart below summarizes the logical flow executed inside the main control loop of the Arduino sketch.



6. Arduino Program

The complete Arduino sketch used for this project -covering sensor initialization, the lid control routine, wet/dry sorting logic, fill-level computation, buzzer alerting, and OLED status rendering -is listed below.

```

/*
=====
SMART DUSTBIN - Wet/Dry Waste Segregation with OLED Display
=====
  
```

Hardware:

- Arduino UNO
- 3x HC-SR04 Ultrasonic Sensors
- 2x Servo Motors
- Soil Moisture Sensor
- Piezo Speaker (Buzzer)
- OLED SSD1306 128x64 (I2C)

NOTE: Adjust pin numbers below if your actual wiring differs.

```
=====
*/

#include <Servo.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// ----- OLED CONFIG -----
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

// ----- PIN DEFINITIONS -----
// Ultrasonic 1 - Lid
const int trigLid = 6;
const int echoLid = 7;

// Ultrasonic 2 - Waste detection (wet/dry)
const int trigWaste = 8;
const int echoWaste = 9;

// Ultrasonic 3 - Bin fill level
const int trigFill = 10;
const int echoFill = 12;

// Servos
const int servoLidPin = 3;
const int servoFlapPin = 5;

// Soil moisture sensor (analog)
const int soilPin = A0;

// Buzzer
const int buzzerPin = 11;

Servo lidServo;
Servo flapServo;

// ----- THRESHOLDS (TUNE THESE) -----
const int LID_OPEN_DIST_CM = 15; // hand distance to trigger lid opening
const int WASTE_DROP_DIST_CM = 10; // distance to detect waste has been dropped
const int BIN_HEIGHT_CM = 30; // total height of bin from sensor to bottom
const int SOIL_WET_THRESHOLD = 500; // below this analog value = wet
const int FULL_PERCENT_LIMIT = 90; // buzzer triggers at this fill %

// ----- SERVO POSITIONS (centered at 90, swings +-90) -----
const int LID_CLOSED_ANGLE = 0; // rest position
const int LID_OPEN_ANGLE = 90; // +90 from rest

const int FLAP_DRY_ANGLE = 0; // +90 from rest
const int FLAP_WET_ANGLE = 180; // -90 from rest

// ----- STATE VARIABLES -----
```

```

bool lidOpen = false;
String lastWasteType = "---";
int fillPercent = 0;

// ----- FUNCTION: Get distance from HC-SR04 -----
long getDistanceCM(int trigPin, int echoPin) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH, 30000); // 30ms timeout (~5m range)
  if (duration == 0) return -1; // no echo received

  long distance = duration * 0.034 / 2; // convert to cm
  return distance;
}

void setup() {
  Serial.begin(9600);

  // Ultrasonic pins
  pinMode(trigLid, OUTPUT);
  pinMode(echoLid, INPUT);
  pinMode(trigWaste, OUTPUT);
  pinMode(echoWaste, INPUT);
  pinMode(trigFill, OUTPUT);
  pinMode(echoFill, INPUT);

  // Buzzer
  pinMode(buzzerPin, OUTPUT);
  digitalWrite(buzzerPin, LOW);

  // Servos
  lidServo.attach(servoLidPin);
  flapServo.attach(servoFlapPin);
  lidServo.write(LID_CLOSED_ANGLE);
  flapServo.write(FLAP_DRY_ANGLE);

  // OLED init
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    while (true); // halt if OLED not found
  }
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(0, 0);
  display.println("Smart Dustbin");
  display.println("Initializing...");
  display.display();
  delay(1500);
}

void loop() {
  // ----- 1. LID CONTROL -----
  long lidDist = getDistanceCM(trigLid, echoLid);

  if (lidDist > 0 && lidDist <= LID_OPEN_DIST_CM) {
    if (!lidOpen) {
      lidServo.write(LID_OPEN_ANGLE);
      lidOpen = true;
    }
  }
  else {
    if (lidOpen) {
      delay(2000); // keep lid open briefly before closing
    }
  }
}

```

```

    lidServo.write(LID_CLOSED_ANGLE);
    lidOpen = false;
  }
}

// ----- 2. WASTE DETECTION + WET/DRY SORTING -----
long wasteDist = getDistanceCM(trigWaste, echoWaste);

if (wasteDist > 0 && wasteDist <= WASTE_DROP_DIST_CM) {
  int soilValue = analogRead(soilPin);

  if (soilValue < SOIL_WET_THRESHOLD) {
    flapServo.write(FLAP_WET_ANGLE);
    lastWasteType = "WET";
  } else {
    flapServo.write(FLAP_DRY_ANGLE);
    lastWasteType = "DRY";
  }

  delay(1500); // allow waste to fall through before resetting flap
  flapServo.write(FLAP_DRY_ANGLE); // return to rest position
}

// ----- 3. BIN FILL LEVEL -----
long fillDist = getDistanceCM(trigFill, echoFill);

if (fillDist > 0) {
  fillPercent = map(fillDist, BIN_HEIGHT_CM, 0, 0, 100);
  fillPercent = constrain(fillPercent, 0, 100);
}

bool binFull = (fillPercent >= FULL_PERCENT_LIMIT);
digitalWrite(buzzerPin, binFull ? HIGH : LOW);

// ----- 4. OLED DISPLAY UPDATE -----
display.clearDisplay();
display.setCursor(0, 0);
display.setTextSize(1);
display.println("== Smart Dustbin ==");

display.setCursor(0, 14);
display.print("Lid: ");
display.println(lidOpen ? "OPEN" : "CLOSED");

display.setCursor(0, 26);
display.print("Waste: ");
display.println(lastWasteType);

display.setCursor(0, 38);
display.print("Fill: ");
display.print(fillPercent);
display.println("%");

display.setCursor(0, 52);
if (binFull) {
  display.println("!! BIN FULL !!");
} else {
  display.println("Status: Normal");
}

display.display();

delay(200);
}

```

7. Simulation Results

The complete circuit was built and tested in the Circuit design circuit simulator prior to physical implementation. This allowed every operating state of the system -lid closed, lid open, dry-waste sorting, wet-waste sorting, and the bin-full alert -to be exercised and visually confirmed on the OLED output before committing to hardware. The simulation screenshots below capture each of these states.

7.1 Idle State -Lid Closed

In the resting state, the lid is closed, no waste has been detected, and the bin is empty. The OLED confirms Lid: CLOSED, Waste: ---, Fill: 0%, and Status: Normal.

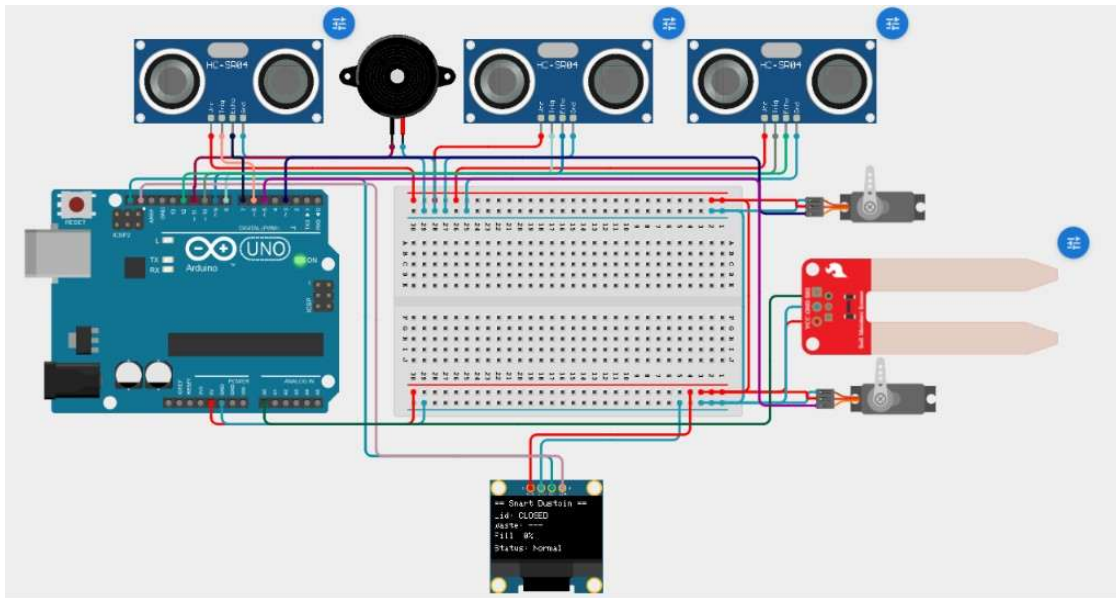


Fig. 7.1 -Idle state: lid closed, bin empty, status normal

7.2 Lid Open -Awaiting Waste

When an object is detected near the lid sensor, the lid servo opens the lid immediately. No waste has yet been classified, so the waste-type field remains blank while the lid state updates to OPEN.

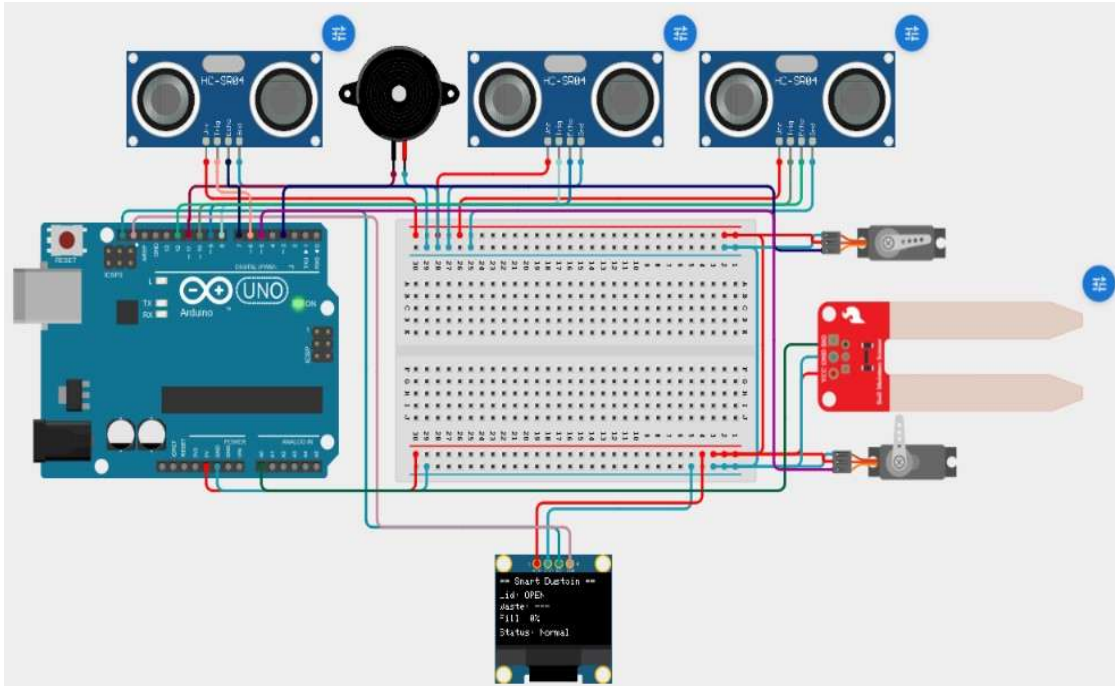


Fig. 7.2 -Lid open, no waste detected yet

7.3 Wet Waste Detected

With the lid open, a waste item is passed through the drop-detection sensor while the soil moisture sensor reads a value below the configured wet threshold. The flap servo routes the item to the wet compartment, and the display updates to Waste: WET.

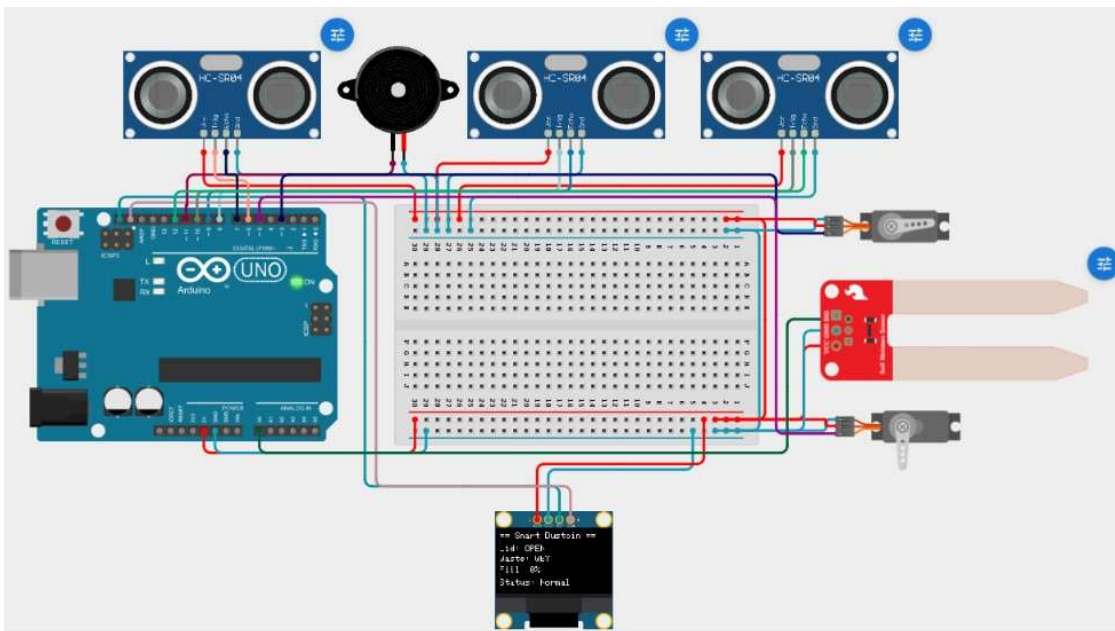


Fig. 7.3 -Wet waste detected and classified

7.4 Dry Waste Detected

Repeating the same sequence with a moisture reading above the wet threshold causes the flap to route the item to the dry compartment instead, with the display updating to Waste: DRY.

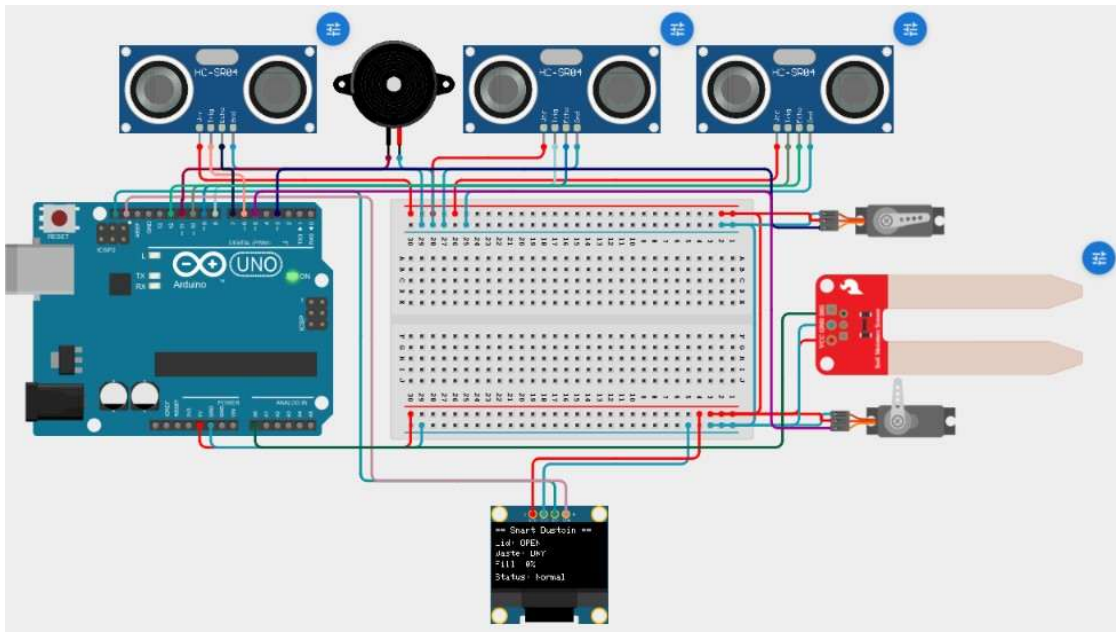


Fig. 7.4 -Dry waste detected and classified

7.5 Bin-Full Alert (Lid Open)

As the fill-level sensor reports a decreasing distance to the waste surface (simulating an accumulating bin), the computed fill percentage rises. Once it reaches the 90% threshold, the buzzer is switched on and the display shows the BIN FULL warning, even while the lid remains open.

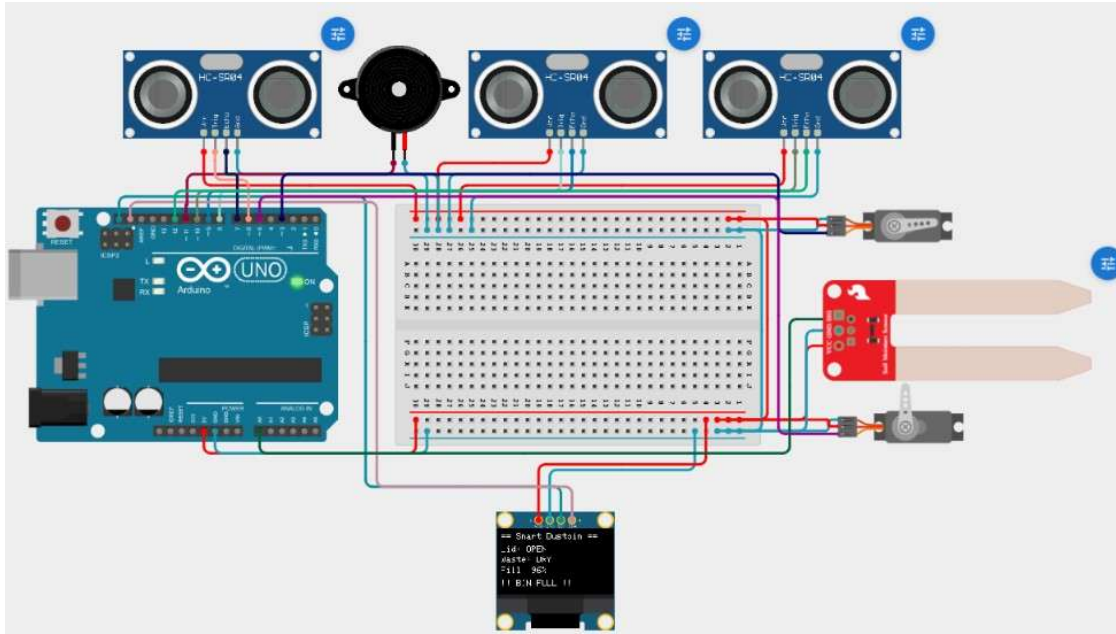


Fig. 7.5 -Fill level at 96%, buzzer active, bin-full warning shown

7.6 Bin-Full Alert (Lid Closed)

The fill-level monitoring and buzzer alert operate independently of the lid state -the alert persists correctly even after the lid has returned to its closed position, confirming that the fill-tracking logic is not reset or interrupted by lid activity.

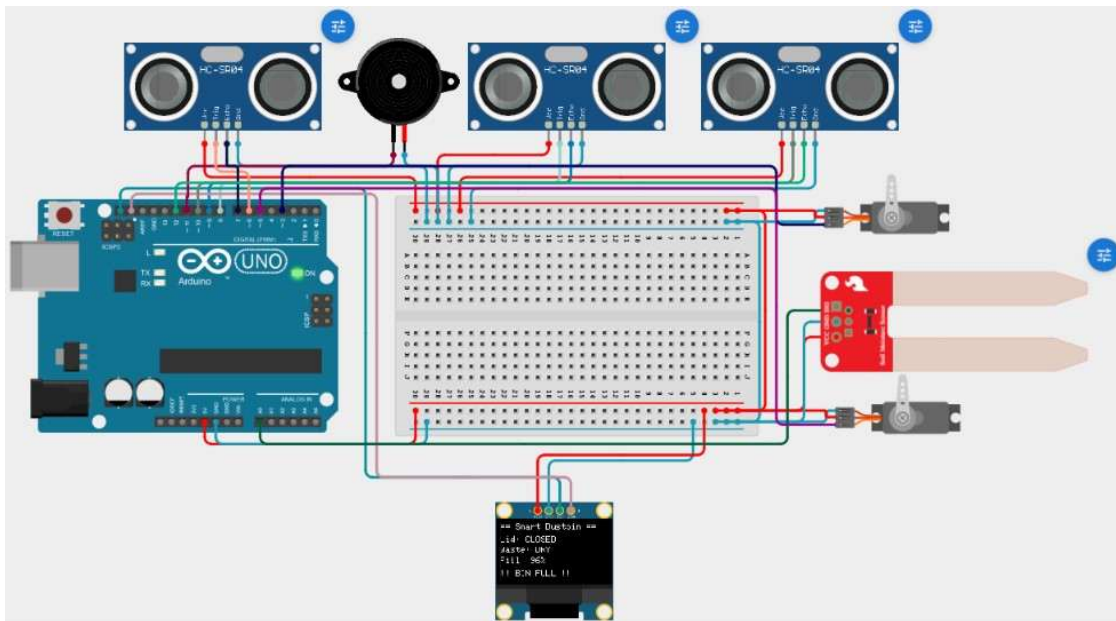


Fig. 7.6 -Bin-full alert persists correctly with the lid closed

7.7 Breadboard Wiring View

The following view shows the physical layout of the simulated breadboard wiring, corresponding directly to the schematic in Section 2.

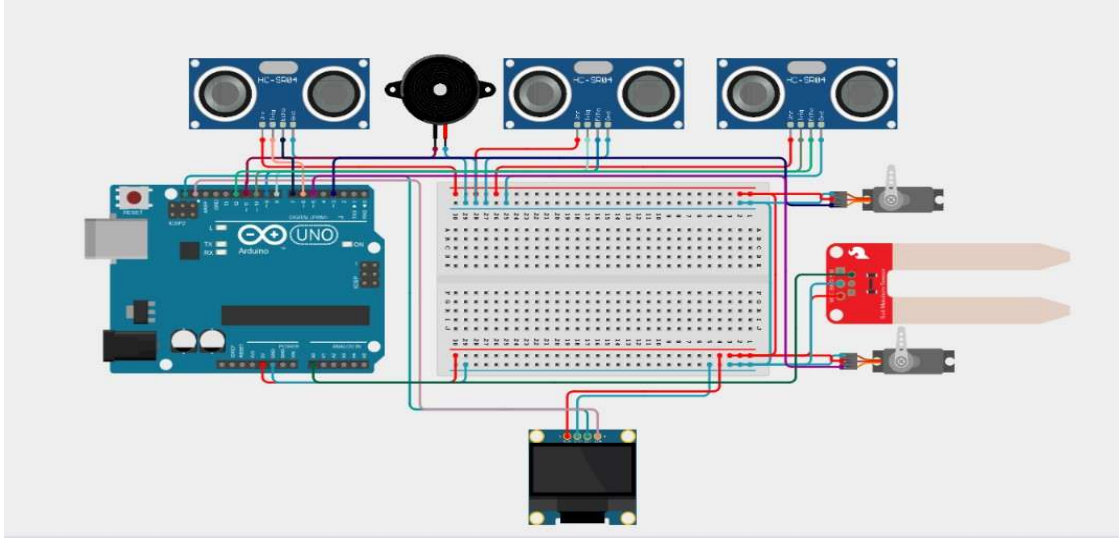


Fig. 7.7 -Breadboard-level wiring of all modules

7.8 Observations

- The lid responded correctly and immediately to objects within the configured detection range, with no false triggers observed during repeated simulation runs.
- The wet/dry classification correctly followed the soil moisture sensor's analog reading in every test case, confirming the threshold value chosen was appropriate for the simulated inputs.
- The fill-percentage calculation tracked the simulated distance changes linearly and triggered the buzzer precisely at the configured 90% threshold.
- The OLED display updated all four status fields consistently and without flicker or lag across every tested state.

8. Advantages

- Touchless lid operation improves hygiene by eliminating the need to physically touch the bin to dispose of waste.
- Automatic wet/dry segregation at the point of disposal reduces contamination of recyclable material and simplifies downstream waste processing.
- Real-time fill-level monitoring with an audible alert prevents overflow and enables more efficient, need-based collection scheduling.
- The OLED status display gives immediate, at-a-glance visibility into the bin's condition without requiring the lid to be opened for inspection.
- Built entirely from low-cost, widely available components (Arduino Uno, HC-SR04 sensors, hobby servos), keeping the overall cost of the system low.
- The modular sensor/actuator design makes the system easy to extend with additional features such as wireless reporting or additional waste categories.

9. Limitations

- The soil moisture sensor is a proxy for wet/dry classification and may misclassify borderline items (e.g., slightly damp paper) that do not clearly fall into either category.

- Ultrasonic sensors can give inconsistent readings with soft, sound-absorbing, or irregularly shaped waste items, occasionally affecting the fill-level accuracy.
- The current design has been validated only in simulation; a physical prototype would need to account for real-world factors such as sensor misalignment, servo torque under load, and enclosure design.
- The system has no wireless connectivity in its current form, so fill-level status can only be viewed locally on the OLED display.
- Powering multiple servos directly from the Arduino's 5V rail is not recommended for a physical build, as noted in the schematic; an external supply and appropriate current handling would be required.
- The bin-height and threshold constants are hard-coded and would need recalibration for bins of different physical dimensions.

10. Applications

- Household and residential waste bins, where automatic segregation reduces the burden on residents to sort waste manually.
- Office buildings, schools, and college campuses, where centralized smart bins can reduce janitorial workload and improve recycling rates.
- Public spaces such as parks, railway stations, and bus stands, where fill-level monitoring can optimize municipal waste-collection routes.
- Hospitals and healthcare facilities, where touchless operation directly supports hygiene and infection-control goals.
- As a building block for larger smart-city waste management infrastructure, where multiple bins report status to a central collection-scheduling system.

11. Future Scope

- Add a Wi-Fi or GSM module (e.g., ESP8266/ESP32) to transmit fill-level and waste-classification data to a cloud dashboard or mobile app for remote monitoring.
- Introduce a third waste category (e.g., e-waste or hazardous waste) with an additional sensor and sorting compartment.
- Replace the soil moisture sensor with a more advanced sensing approach, such as capacitive or infrared-based material classification, to improve sorting accuracy.
- Add solar-panel charging and a rechargeable battery to make the unit fully self-powered and suitable for outdoor deployment.
- Integrate GPS and route-optimization logic across a network of bins to help municipal services plan efficient collection routes based on real-time fill data.
- Design and fabricate a weatherproof physical enclosure suitable for outdoor or high-traffic public deployment.

12. Conclusion

The Smart Dustbin project successfully demonstrates how a small set of low-cost sensors and actuators, coordinated by a single Arduino Uno, can automate three distinct and genuinely useful waste-management tasks: touchless lid access, automatic wet/dry waste segregation, and real-time fill-level monitoring with alerting. The complete system was designed, wired, and validated through circuit simulation in Circuit design , with every operating state -from idle to full-bin alert -verified against the expected behavior described in the working principle.

Beyond its immediate function, this project provided practical, hands-on experience in coordinating multiple ultrasonic sensors, servo actuators, an analog sensor, and an I2C display within a single real-time control loop, including the timing and threshold-tuning challenges that come with making several subsystems operate reliably together. With the extensions outlined in the future scope -particularly wireless connectivity and a physical enclosure -this design provides a solid, well-validated foundation for a deployable smart waste-management device.